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003401

May 2024

B.Tech. (CE/CE(HINDI)/IT/CSE(AIML))

(Fourth Semester)

Discrete Mathematics (PCC-CS-401)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Define Well Ordering Principle. 1.5

(b) Three red balls are to be placed in ten numbered boxes but one box can contain exactly one ball. Find the number of distinct ways in which the balls can be placed. 1.5

(c) Every student of a class participates in either one or two or all the three games being played in the school. Make the Venn diagram for this. 1.5

(d) Define monoid in relevance to Algebraic Structure. 1.5

(e) What do you mean by Non-planar Graph ? 1.5

(f) What do you mean by Contradiction ? 1.5

(g) What do you mean by Weighted Graph ? 1.5

(h) In a group of 45 people, how many minimum number of people have their birthday in the same month ? 1.5

(i) What do you mean by Ternary Relation ? 1.5

(j) What do you mean by one-to-one and on-to function ? 1.5

Part B

2. (a) If $f(x)$ and $g(x)$ be the function from the set of integers defined as $f(x) = 2x^2 + 2$ and $g(x) = 3x^2 + 3$ then determine yjr following : **10**

(i) $f \circ g$

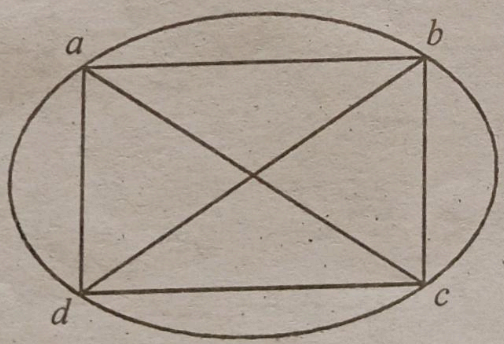
(ii) $g \circ f$.

(b) If $f(x)$ be a function from the set $\{1, 2, 3, 4\}$ to the set $\{a, b, c, d\}$ with $f(1) = a$, $f(2) = b$, $f(3) = c$, $f(4) = d$, then determine $f^{-}(a), f^{-}(b), f^{-}(c), f^{-}(d)$. 5

3. (a) Consider three propositions- (i) The temperature exceeds 70 degree celcius (ii) The alarm will be sounded. (iii) If the temperature exceeds 70 degree celcius then The alarm will be sounded. 5

In what situation propositions 3 will be true ?

(b) Determine the chromatic number for the given graph : 10



4. Obtain the, both, CNF and DNF for the given Boolean Function- $(y + x')z$. 15

5. (a) If R is a binary relation on the set of integers i. e. $\{1, 2, 3, ...\}$ such that $(a, b) \in R$ if and only if $(a - b) \geq 1)$ then determine whether the relation R is : 10

(i) Reflexive

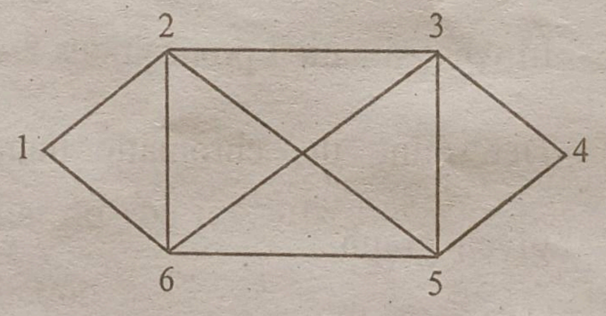
(ii) Symmetric

(iii) Antisymmetric

(iv) Transitive.

(b) Define Partial Ordering Relation and Equivalence Relation. 5

6. (a) Find the Eulerian path in the given graph. 5



(b) Prove that some of the infinite sets are not countable. 10

7. Write short notes on the following : 15

(a) Monoid, Group and Semi-group

(b) Give example of algebraic system with two Binary operations

(c) Give an example of Infinite Set.

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003402

May 2024

B. Tech. (CE/CE(HINDI)/IT/CSE(AIML))

(Fourth Semester)

Computer Organization and Architecture

(PCC-CS-402)

Time : 3 *Hours*] [*Maximum Marks* : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) What do you mean by stored program control concept ? **1.5**

   (b) What is program status word ? **1.5**

   (c) Explain the concept of Cache Coherence. **1.5**

   (d) Differentiate between computer architecture and computer organization. **1.5**

   (e) Draw the flowchart explaining the process of non-restoring division algorithm. **1.5**

(f) Bring out the differences between 8085 and 8086 microprocessors. 1.5

(g) Briefly explain the different I/O interfaces - PCI, SCSI, USB. 1.5

(h) How can we calculate the speedup and throughput of a system ? 1.5

(i) An instruction is stored at location 200 with address field located at 201. The value of address field is 320. A processor register R1 contains the value 500. Calculate the effective address if the addressing mode is : 1.5

 (i) Direct

 (ii) Relative

 (iii) Register indirect

(j) What is memory interleaving ? How is it useful ? 1.5

Part B

2. (a) Explain Flynn's classification of parallel processors. 5

 (b) Represent the decimal number $(-262.125)_{10}$ in single precision floating point format. 5

 (c) Explain the ripple carry adder/subtractor using the circuit diagram. 5

3. (a) Explain stack based CPU organization. Use a suitable example to demonstrate the types of instruction formats used in this type of organization ? 10

(b) Obtain the result of multiplying $(-6)_{10}$ and $(-9)_{10}$ using booth's multiplier. Draw the flowchart to justify the steps used in obtaining the result. 5

4. (a) What is asynchronous data transfer ? Explain any *one* method used in asynchronous data transfer in detail. 5

(b) What is micro-programmed control unit ? How to obtain address sequencing ? 10

5. (a) Explain the 16-bit status/flag register in 8086 microprocessor. If an addition operation is performed on two values – 81 and FE (hexadecimal), what is the resultant value of this register ? 5

(b) What are the various pipeline hazards that are likely to occur in computer architecture ? 10

6. (a) A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit

P.T.O.

words. Calculate how many bits are required for addressing the main memory ? Also, how many bits are needed to represent the TAG, SET and WORD fields ? 5

(b) Explain the working of a DMA controller with the help of block diagram. What are the various modes of transfer used by DMA ? 10

7. Write short notes on the following : 15

(a) Priority Interrupts

(b) Write-back and Write-through policies

(c) RISC *vs*. CISC instruction sets

(d) Hardware interrupts

(e) Hierarchical memory organization

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011403

May 2024

B. Tech. (IT) (Fourth Semester)

Organizational Behaviour (HSMC-03)

Time : 3 Hours] [*Maximum Marks* : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. Write short notes on the following :

(a) Some ethical dilemmas managers may face. **1.5**

(b) Role of MIS to support decision-making and organizational performance **1.5**

(c) Perceptual errors **1.5**

(d) Elements of organizational culture **1.5**

(e) Determinants of personality **1.5**

(f) Organizational interventions in managing stress **1.5**

(g) Sources of power 1.5

(h) Group Cohesiveness and its impact on group performance. 1.5

(i) Differentiate between matrix and network organization design. 1.5

(j) Role of OD in facilitating orgnaisational change. 1.5

Part B

2. How has the management concept evolved from traditional viewpoints to contemporary perspectives such as contingency and quality management ? 15

3. Examine motivational theories such as Maslow's hierarchy of needs, Herzberg's two-factor theory, and expectancy theory. How do these theories apply to real-world scenarios ? 15

4. Discuss techniques and strategies for managing conflict within teams. How can constructive conflict resolution contribute to team effectiveness ? 15

5. Outline the functions of leadership. What are different leadership styles and how do they impact performance ? 15

6. Discuss the reasons why organisations change. Identify sources of resistance to change within organizations and discuss strategies for managing resistance effectively. 15

7. (a) Define organizational climate and discuss its influence on employee attitudes, behaviors, and job satisfaction. 10

   (b) Discuss the contemporary challenges faced by OB specialists in a dynamic and globalised workplace. 5

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003404

May 2024

B. Tech. (CE/CE(HINDI)/IT/CSE(AIML)

(Fourth Semester)

Desgin and Analysis of Algorithms

(PCC-CS-404)

Time : 3 *Hours*] [*Maximum Marks* : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Write the recurrence relation for n-ary search and why is it not preferred ? **1.5**

(b) Write the following functions in the increasing order of asymptotic complexity :

$$f1(n) = 2^n,\ f2(n) = n^{(3/2)},$$

$$f3(n) = n * \log(n),\ f4(n) = n^{\log(n)}.$$ **1.5**

(c) Why Greedy method is not suggested for 0/1 Knapsack problem ? 1.5

(d) Bellman-Ford Algorithm always find negative weight cycle in the graph. Justify your answer. 1.5

(e) In which scenario a sorting technique is called stable ? Also give example of 2 stable sorting techniques. 1.5

(f) Merge the following files optimally :

5, 9, 4, 2, 1, 10. 1.5

(g) Explain the significance of asymptotic notations. 1.5

(h) Explain explicit and implicit constraints for Hamiltonian Cycle. 1.5

(i) Describe Least Cost Search function in brief. 1.5

(j) Differentiate P, NP-Hard and NP-Complete problems. 1.5

Part B

2. (a) Solve the following recurrence relations : 5

(i) $T(n) = T(\sqrt{n}) + n$

(ii) $T(n) = 7T\,\mathrm{Floor}(n/3) + n^2$.

(b) Write the algorithm for divide and conquer based sorting technique which runs with same time complexity in all scenarios and also compute the time complexity of the algorithm. 10

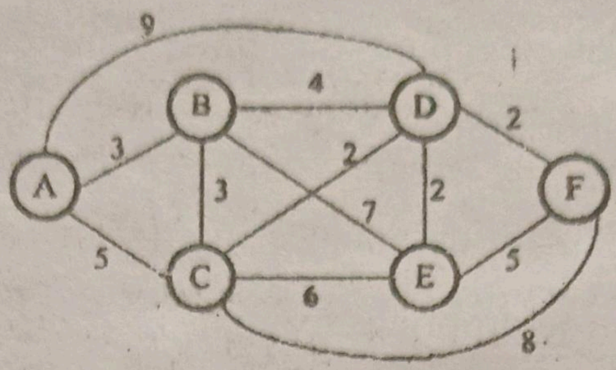
3. (a) Explain dominance rule (Merge-Purge) with an appropriate example. 5

(b) Differentiate backtracking and dynamic programming algorithm design techniques corresponding to their application domain. Also explain the N-Queen problem in detail using backtracking. 10

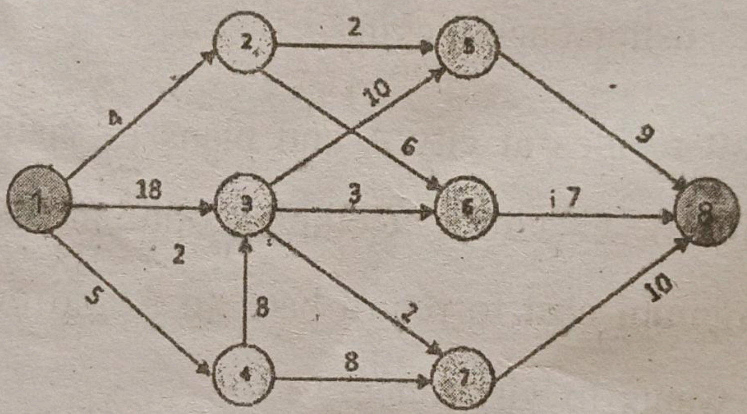
4. Amit resides at city A and plans to visit various cities B, C and D. Compute optimal route with minimum cost to return back to A and every city he wants to visit once. 15

	A	B	C	D
A	0	8	4	12
B	4	0	6	14
C	6	8	0	7
D	9	4	5	0

5. (a) Compute the Shortest path from A to F using greedy method : 5



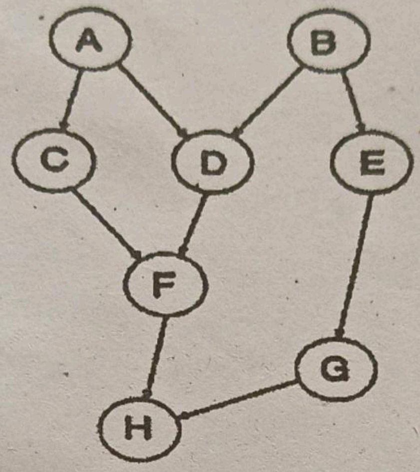
(b) Write Ford Fulkerson-Network Flow algorithm and compute the maximum unit flow from node 1 to node 8. 10



6. (a) Solve the knapsack problem using Branch and bound technique :

m = 30, (w1, w2, w3, w4) = (18, 15, 8, 12), (p1, p2, p3, p4) = (36, 24, 8, 12). 10

(b) If A to H are different steps of a task and their dependency is given in the graph below. Compute topological order of events for successfully completion of task : 5



7. (a) Explain any *two* types of Randomized algorithm techniques with corresponding applications and scope. 8

(b) Explain the utility of Approximation algorithms with appropriate example. 7